

B.Tech /ALL /ODD/SEM_1/M 101/2021-2022

PAPER TYPE: REGULAR
YEAR: 2022

MATHEMATICS I M 101

TIME ALLOTTED: 3 HOURS

FULL MARKS: 70

*The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable*

GROUP – A (Multiple Choice Type Questions)

1. Answer any **ten** from the following, choosing the correct alternative of each question: **10×1=10**

SL. NO.	Question	Marks	CO
(i)	If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, then $\text{curl } \vec{r}$ is a) $\vec{i}$ b) $\vec{j}$ c) $\vec{k}$ d) $\vec{0}$.	1	CO1
(ii)	The pair of the functions which satisfy the conditions of Cauchy's M.V.T. in $[-1,1]$ is a) $\sqrt[3]{x}, x^2$ b) $\frac{1}{\sqrt{x}}, \sqrt{x}$ c) $ x , x$ d) $\sqrt[3]{x+2}, x^2$	1	CO3
(iii)	$\sin^{-1} \frac{x^3 + y^3}{x + y}$ is a a) homogeneous function of degree 2 b) homogeneous function of degree 1 c) homogeneous function of degree $\frac{1}{2}$ d) non-homogeneous function.	1	CO3
(iv)	The value of a for which $2x\hat{i} + (2x^2 + y)\hat{j} + (3az + 2x^3)\hat{k}$ is a solenoidal is a) -1 b) 0 c) 1 d) 2.	1	CO2
(v)	If $\lambda^3 - 6\lambda^2 + 9\lambda = 4$ is the characteristic equation of a square matrix A, then A^{-1} is a) $A^2 - 6A + 9I$ b) $\frac{1}{4}A^2 - \frac{3}{2}A + \frac{9}{4}I$ c) $\frac{1}{4}A^2 - \frac{3}{2}A + \frac{9}{4}$	1	CO1

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT

d) $A^2 - 6A + 9$

- | | | | |
|--------|---|---|-----|
| (vi) | If Rank(A)=, then the matrix $A_{3 \times 3}$ is
a) singular
b) non singular
c) orthogonal
d) None | 1 | CO2 |
| (vii) | Identify the correct statements
a) A bounded monotonic sequence is divergent
b) A bounded monotonic sequence is convergent
c) A bounded monotonic sequence is oscillatory
d) A bounded monotonic sequence may not have finite limit | 1 | CO1 |
| (viii) | Which of the following theorem can be applied on $f(x)= x $ in the interval $[-1, 1]$

a) Rolle's theorem
b) Lagrange Mean value theorem
c) Cauchy Mean value theorem
d) None of these | 1 | CO2 |
| (ix) | $\int_0^1 \int_y^{\sqrt{y}} dx dy =$

a) 1/2
b) 1/6
c) 2/3
d) 4/3 | 1 | CO4 |
| (x) | The series $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ is
a) absolutely convergent
b) conditionally convergent
c) oscillatory
d) divergent | 1 | CO2 |
| (xi) | If $A = \begin{pmatrix} 5 & 2 & 3 \\ 1 & 0 & 2 \\ 0 & 4 & 0 \end{pmatrix}$, then the sum of the eigen values of A

is
a) 3
b) 5
c) 6 | 1 | CO2 |

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT

- d) 10
- (xii) The value of $\int_0^2 \int_0^2 \int_0^2 xyz \, dx dy dz$ is 1 CO1
- a) 1/2
b) 1/8
c) 8
d) 16

GROUP – B
(Short Answer Type Questions)

Answer any *three* from the following: **3×5=15**

- | SL. NO. | | Marks | CO |
|---------|---|-------|-----|
| 2. | Use mean value theorem to prove that $\frac{x}{1+x^2} < \tan^{-1} x < x$ when $0 < x < \frac{\pi}{2}$ | 5 | CO3 |
| 3. | Reduce the matrix $A = \begin{bmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{bmatrix}$ to an equivalent Echelon matrix and hence find its rank. | 5 | CO3 |
| 4. | Test the convergence of the series $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$ | 5 | CO2 |
| 5 | Find the eigen values and the corresponding eigen vectors of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ | 5 | CO1 |
| 6. | What is the value of $\int \int y dx dy$ over the region bounded by $y^2 = x$ and $y = x^2$. | 5 | CO2 |

GROUP – C
(Long Answer Type Questions)

Answer any *three* from the following: **3×15=45**

- | SL. NO. | | Marks | CO No. |
|---------|---|-------|--------|
| 7. (a) | Verify Cayley-Hamilton theorem for the matrix | 5 | CO2 |

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT

$$\begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}. \text{ Hence, evaluate } A^{-1}.$$

- (b) Show that $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$ is convergent series. 5 CO3
- (c) Solve by matrix method the system of equations 5 CO1
- $$x+z=0$$
- $$3x+4y+5z=2$$
- $$2x+3y+4z=1.$$
8. (a) Use Cayley-Hamilton theorem to find A^{-1} , where 5 CO1
 $A = \begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix}$
- (b) If $u = u\left(\frac{y-x}{xy}, \frac{z-x}{zx}\right)$ show that 4 CO3
 $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$
- (c) State Lagrange's Mean Value Theorem. Write Taylor's formula for the function 6 CO4
 $f(x) = \log(1+x), -1 < x < \infty$ about $x=2$ with Lagrange's form of remainder after 3 terms.
9. (a) Show that $\vec{A} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$ is 6 CO4
 irrotational. Find the scalar ϕ such that $\vec{A} = \vec{\nabla}\phi$.
- (b) Show that $\text{curl grad } f = \vec{0}$, where $f = x^2y + 2xy + z^2$. 4 CO2
- (c) Find the directional derivative of $f(x, y, z) = x^2yz + 4xz^2$ 5 CO2
 at the point $(1, -2, -1)$ in the direction of the vector $2\vec{i} - \vec{j} - 2\vec{k}$
10. (a) Verify the divergence theorem for $F(x, y, z) = xi + yj + 0k$ and 8 CO3
 the surface $x^2 + y^2 + z^2 = 9$.
- (b) Find the value of $\int_C (3xydx - y^2dy)$ 3 CO2

Where C is the arc of the parabola

$$y = 4x^2 \text{ from } (0,0) \text{ to } (1,4)$$

NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT

- (c) Apply Stoke's theorem to evaluate $\int_C (ydx + zdy + xdz)$, 4 CO3
where C is the curve of intersection of $x^2 + y^2 + z^2 = 9$ and $x + z = 3$.
11. (a) If $u = \sin^{-1} \left\{ \frac{x^{1/3} + y^{1/3}}{\sqrt{x} + \sqrt{y}} \right\}$, then prove that 6 CO3
$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{12} \left(\frac{13}{12} + \frac{\tan^2 u}{12} \right)$$
- (b) Show that $\mathbf{A} = \frac{1}{2} (\hat{i}x^2 + \hat{j}y^2 + \hat{k}z^2)$ is irrotational. 3 CO2
- (c) Evaluate $\iiint_V z(x^2 + y^2) dz dy dx$ where V is the volume of 6 CO4
the cylinder $x^2 + y^2 = 1$ intercepted by the planes $z = 2$ and $z = 3$.

Course Outcomes (COs):

On successful completion of the learning sessions of the course, the learner will be able to

CO1: Recall the properties and formula related to matrix algebra, differential calculus, multivariable calculus, vector calculus and infinite series.

CO2: Determine the solutions of the problems related to matrix algebra, differential calculus, multivariable calculus, vector calculus and infinite series.

CO3: Apply the appropriate mathematical tools of matrix algebra, differential calculus, multivariable calculus, vector calculus and infinite series for the solutions of the problems.

CO4: Analyze different engineering problems linked with matrix algebra, differential calculus, multivariable calculus, vector calculus.